

JKBOSE 2026

CLASS 11TH

Anatomy Of Flowering Plants

LECTURE-03

BOTANY

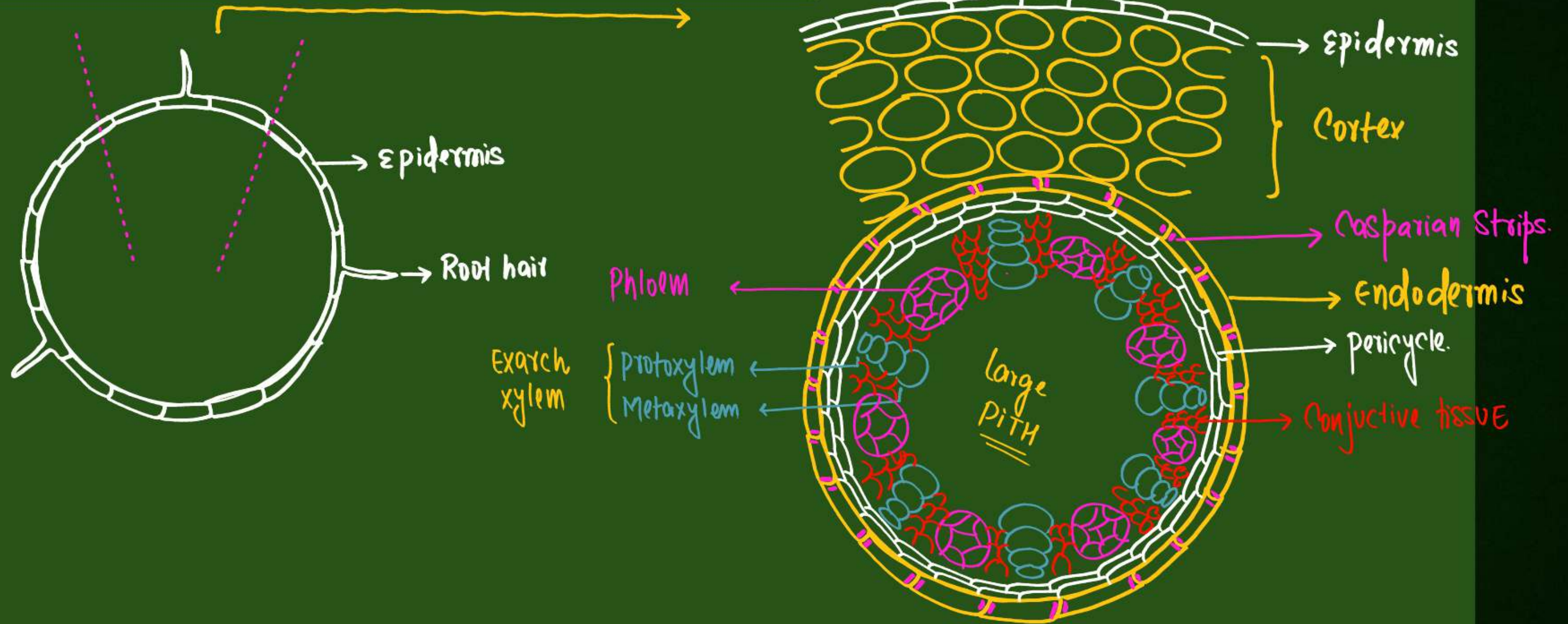
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Topics To Covered

- 1** Anatomy of Monocot Root
- 2** Anatomy of Dicot Stem
- 3** Anatomy of Monocot Stem

ANATOMY OF MONOCOT ROOT

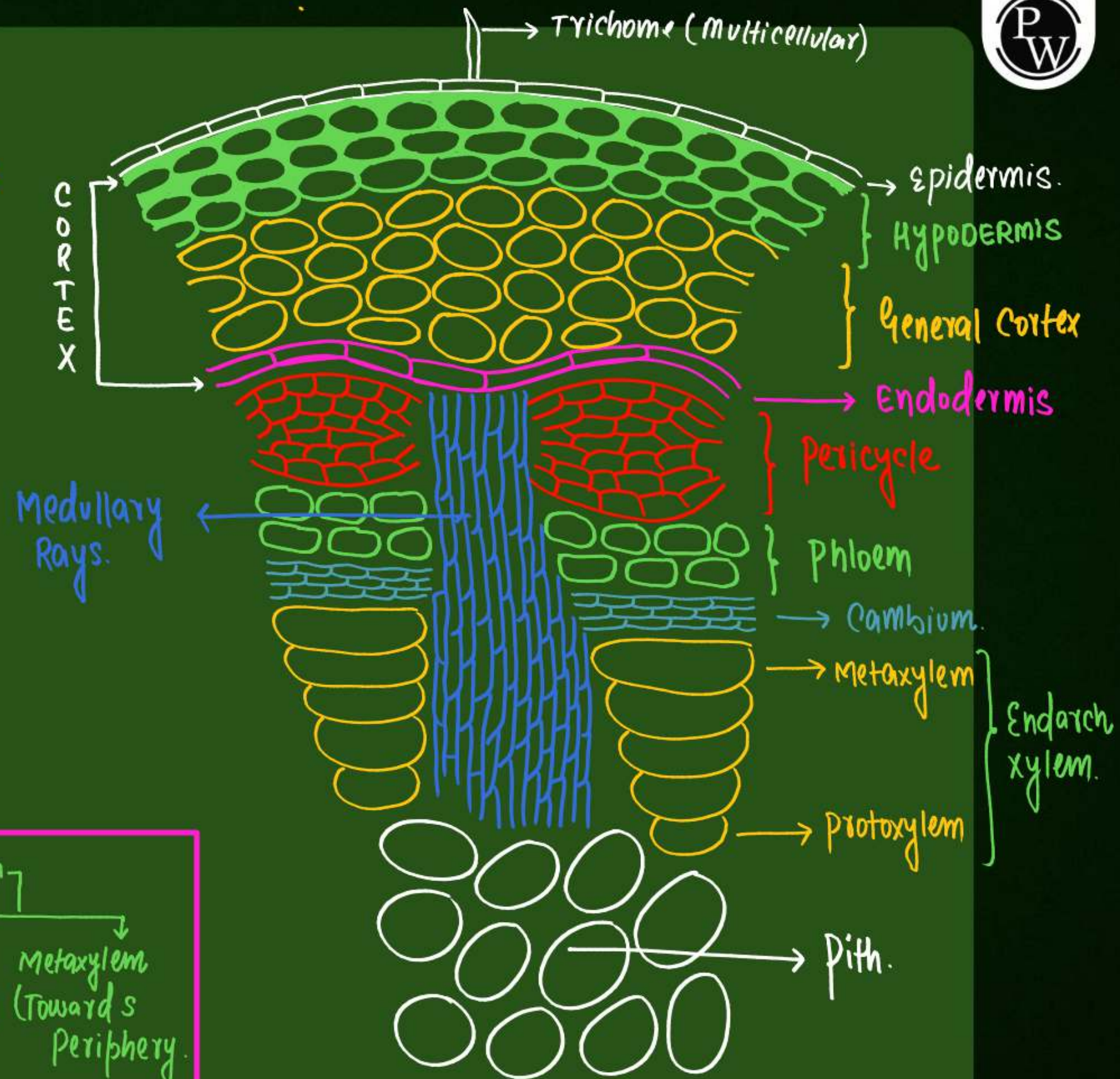
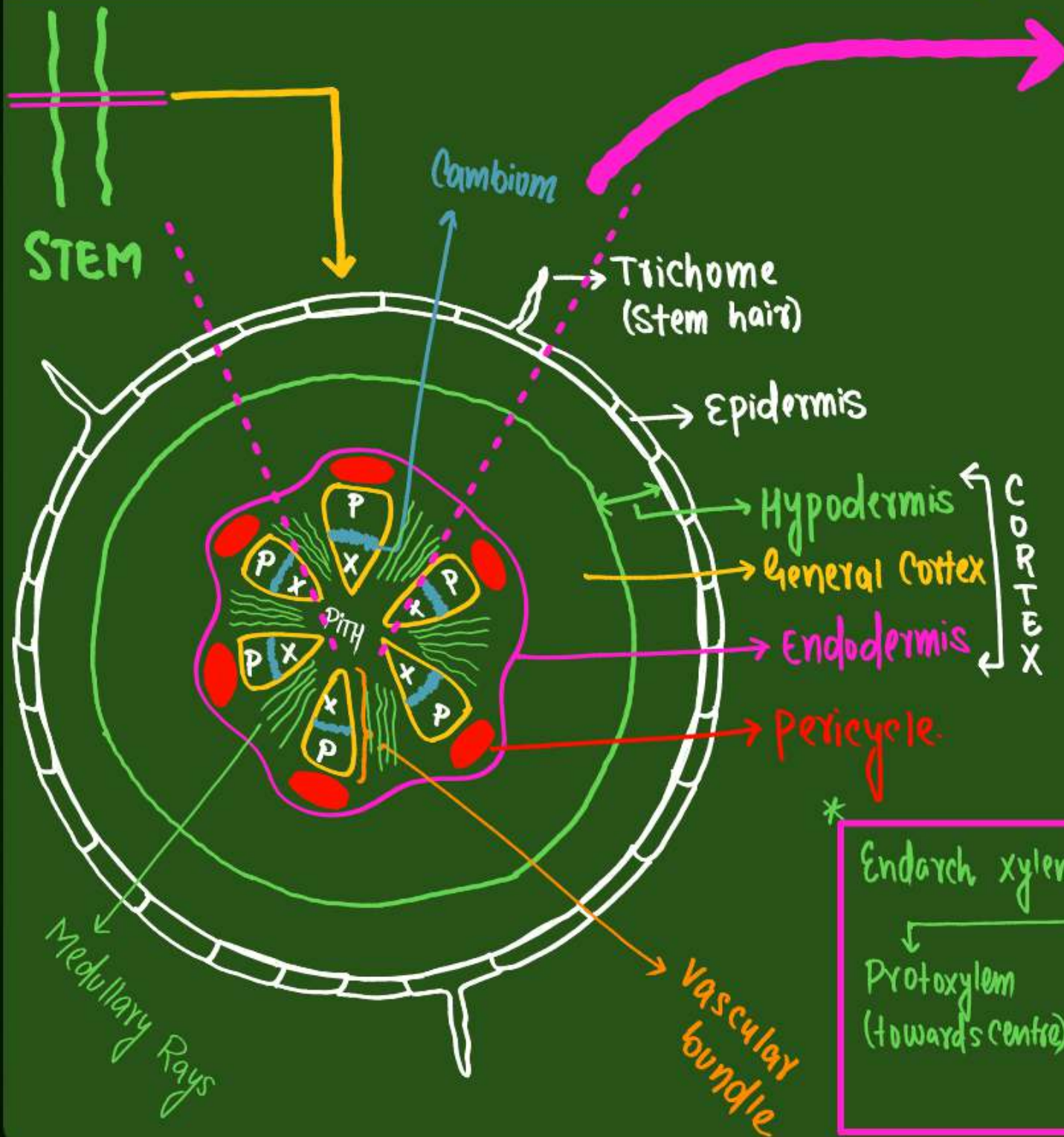


Anatomy of dicot root and monocot root is similar in many aspects except:

1. **VASCULAR BUNDLE:** radial, xylem and phloem patches are more than 6 (polyarch condition).

2. **PITH:** Large and well developed No secondary growth in monocot root.*

Anatomy of Dicot Stem



↪ **EPIDERMIS:** Outer most single layered, protective in nature, bear multicellular trichomes and few stomata.

↪ **CORTEX:** Present between epidermis and pericycle, multilayered, divided into 3 sub-zones:

a) **Hypodermis:** present below epidermis, few layered, collenchymatous in nature, intercellular spaces absent, provide mechanical support to young stem.

b) **General cortex:** present below hypodermis, rounded thin walled and parenchymatous, multilayered, conspicuous intercellular spaces present.

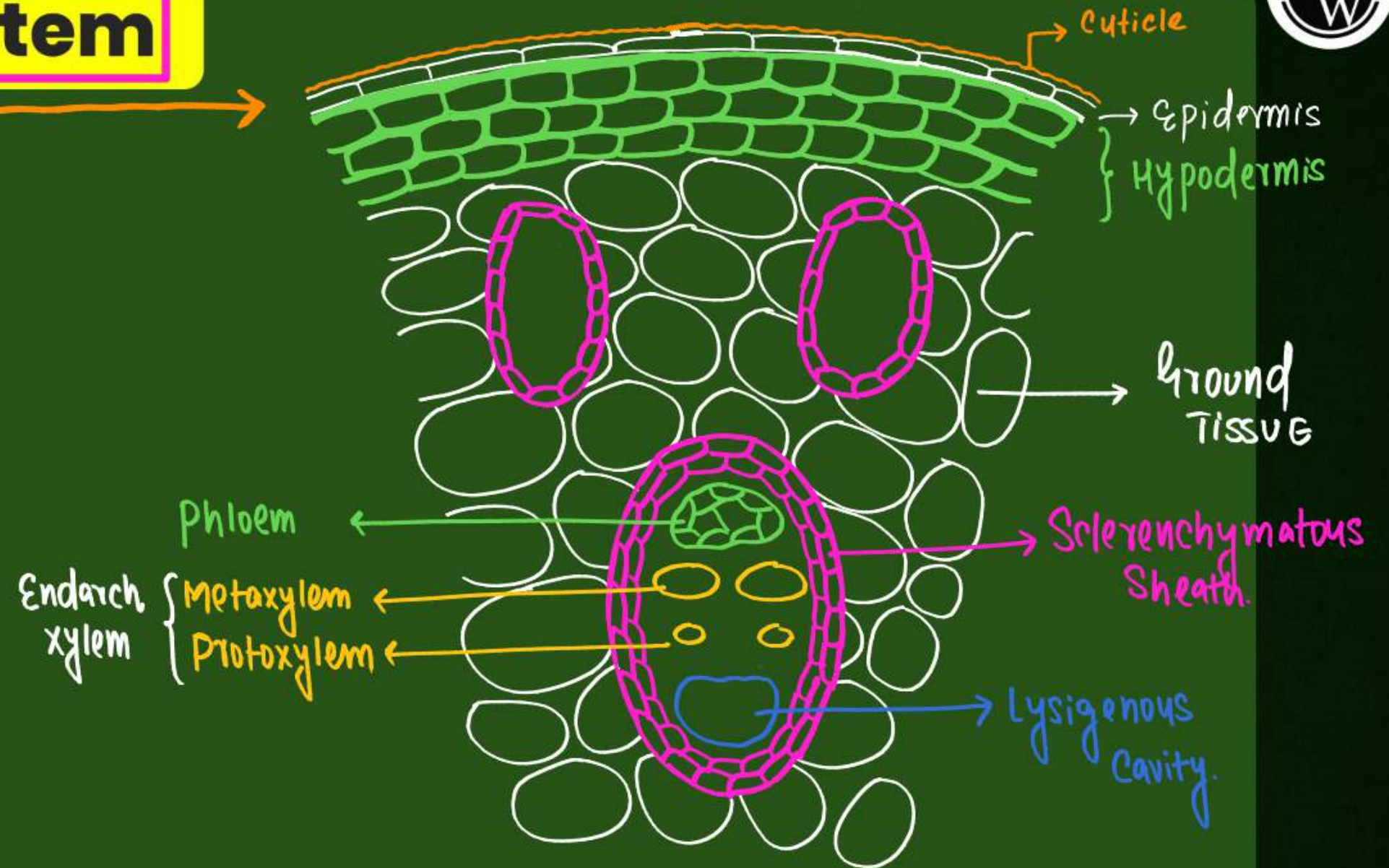
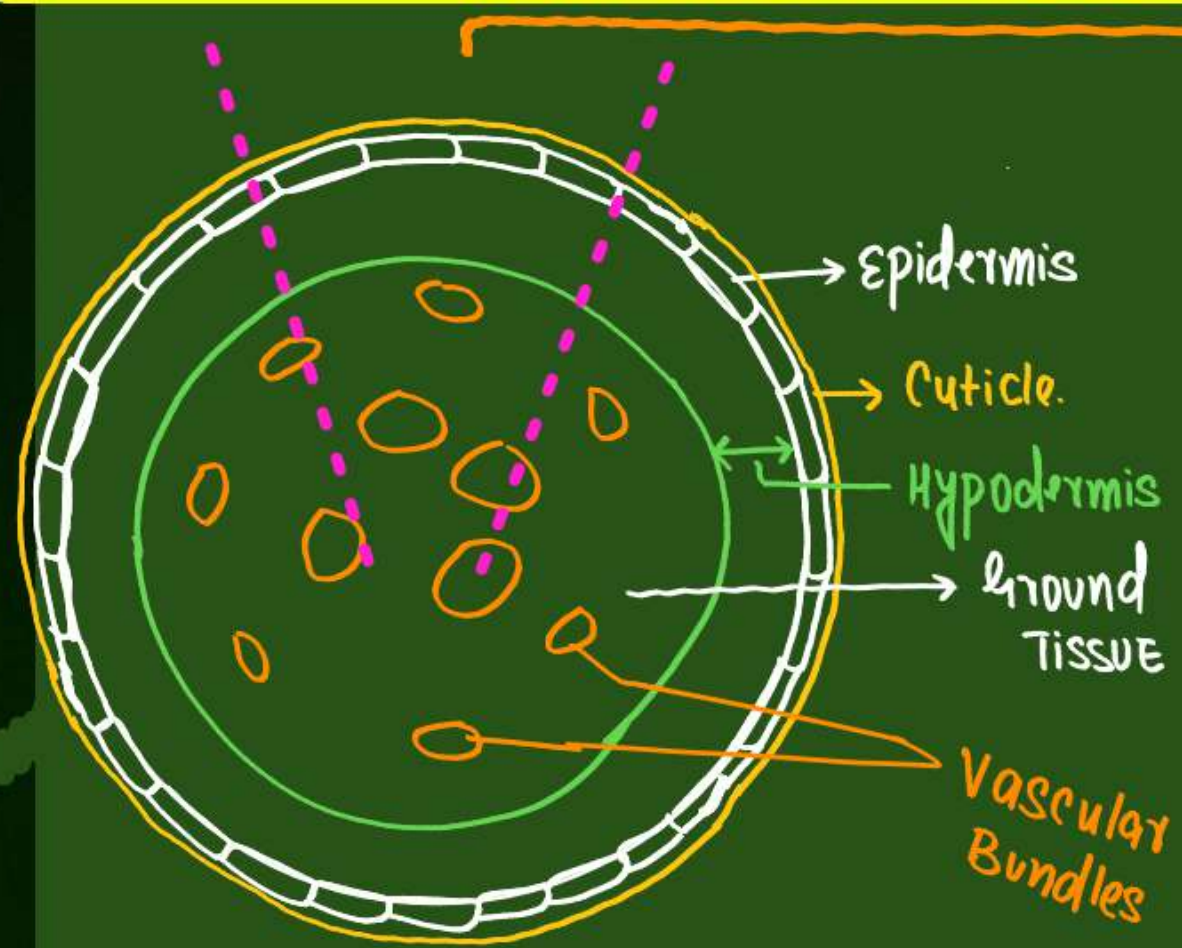
c) **Endodermis:** innermost layer of cortex, single layered, possess starch grains i.e., why it is also called as starch sheath.

↳ **VASCULAR BUNDLE:** large in number, arranged in ring (characteristic feature of dicot stem), vascular bundle is conjoint and open, xylem is endarch.

↳ **MEDULLARY RAYS:** parenchymatous, few layered, present between vascular bundles.

↳ **PITH:** central portion of stem, contains large number of rounded and parenchymatous cells, large intercellular spaces are present.

Anatomy of Monocot Stem



⇒ **EPIDERMIS:** outermost single layered, possess thin waxy layer (cuticle), trichomes are absent.

⇒ **HYPODERMIS:** present inner to epidermis, sclerenchymatous in nature.

⇒ **GROUND TISSUE:** present inner to hypodermis, parenchymatous and large cells.

⇒ **VASCULAR BUNDLE:** present scattered inside ground tissue, abundant in number, small vascular bundles towards periphery and large towards centre, each vascular bundle is surrounded by sclerenchymatous sheath, vascular bundles are conjoint and closed, xylem is endarch, vascular bundles possess lysigenous cavities which stores water, gasses and essential oils.

QUESTIONS

The transverse section of a plant part showed polyarch, radial and exarch xylem, with endodermis and pericycle. The plant part is identified as: (2023 Manipur)

- A** Monocot root
- B** Dicot root
- C** Dicot stem
- D** Monocot stem

QUESTIONS

Casparian strips occur in

(2018)

- A** Epidermis
- B** Pericycle
- C** Cortex
- D** Endodermis

QUESTIONS

H/W

A major characteristic of the monocot root is the presence of:

(2015)

- A** Vasculture without cambium
- B** Cambium sandwiched between phloem and xylem along the radius
- C** Open vascular bundles
- D** Scattered vascular bundles

HOMEWORK

1 Question



Thank
You